Ahmed Elbossily

Contact Information

Ahmed.Elbossily@leuphana.de — Website: AhmedElbossily.github.io

Education

Leuphana University Lüneburg — Lüneburg, Germany
Institute for Production Technology and Systems — 2024–Present

Ph.D. Candidate

- Advisor: Prof. Dr.-Ing. Benjamin Klusemann
- Dissertation Topic: "GPU-Accelerated Smoothed Particle Hydrodynamics for Simulating Solid-State Manufacturing Processes"

Arab Academy for Science, Technology, and Maritime Transport (AASTMT) — Cairo, Egypt
2014–2018

Master of Science (M.Sc.) in Computational Mechanics

- Grade: Excellent
- Advisor: Prof. Mostafa R. A. Atia

Work Experience

Research Assistant & Ph.D. Candidate — Leuphana University
2024–Present

- Conduct research within an ERC-funded project focused on GPU-accelerated solvers for manufacturing-process simulations.
- Develop and optimize high-performance C++/CUDA algorithms for large-scale particle-based simulations on NVIDIA GPUs.
- Profile and analyze CUDA developed solvers using NVIDIA Nsight Systems and Nsight Compute to identify performance bottlenecks, memory inefficiencies, and kernel-level optimization opportunities.
- Validate and test numerical models on NVIDIA GPUs.
- Implement and evaluate multi-GPU solvers using multiple parallel programming models, including OpenMP, MPI, NCCL, and NVSHMEM.
- Developed multi-threaded and multi-process GPU implementations using CUDA-aware MPI, GPUDirect Peer-to-Peer (P2P) communication, Unified Memory, and NCCL.
- Supervise graduation projects focused on GPU programming and parallel computing techniques.

Independent Consultant – Udacity — Udacity – Remote
2018–Present

- Delivere mentoring, live online sessions, project reviews, and technical support to improve learner outcomes.
- Supporte corporate upskilling and accelerator programs, including C++ training for programmers from Stellantis.
- Mentore learners in C++, computer vision, machine learning, and artificial intelligence.
- Edit, review, and audit course materials, including the Autonomous Systems curriculum.

Engineering & Technical Leadership Roles — Egypt, Government Organization
2013–2024

Progressed through engineering, R&D, laboratory-management, advanced-systems, team lead, and technical-leadership positions.

- Develop software tools and engineering solutions for technical and operational challenges.
- Implemente and optimize GPU-accelerated algorithms, focusing on parallel execution and efficient memory access.

- Profile CUDA applications using NVIDIA Nsight Systems and Nsight Compute.
- Plan and execute complex system tests, with a focus on control-system performance and ECU response.
- Analyze test data, diagnose system-level problems, and recommend technically feasible solutions.
- Led multidisciplinary engineering teams and prepared technical documentation, reports, and process instructions.

Teaching Experience

Leuphana University 2024–Present

- Introduction to GPU Programming
- GPU Programming for Business Applications
- C++ for Mechanical Applications
- Mathematics I–II (Exercises)

Refereed Publications

1. **Ahmed Elbossily**, Z. Kallien, R. Chafle, K. A. Fraser, M. Afrasiabi, M. Bambach, and B. Klusemann, "GPU-accelerated meshfree computational framework for modeling the friction surfacing process," *Computational Particle Mechanics*, Vol. 12, No. 5, pp. 3721–3745, 2025. https://doi.org/10.1007/s40571-025-01048-2

2. **Ahmed Elbossily**, Z. Kallien, L. Rath, R. Chafle, M. Afrasiabi, and B. Klusemann, "Experimental–numerical investigation of force-controlled friction extrusion process via feedback-controlled simulation," *Advances in Industrial and Manufacturing Engineering*, Vol. 12, 100184, 2026. https://doi.org/10.1016/j.aime.2026.100184

3. **Ahmed Elbossily**, Z. Kallien, R. Chafle, K. A. Fraser, M. Afrasiabi, and B. Klusemann, "Numerical Investigation of Deposition Efficiency Influencing Factors in the Friction Surfacing Process," *Key Engineering Materials*, Vol. 1050, pp. 1–7, April 2026. https://doi.org/10.4028/p-4tzjoz

Technical Skills

Programming: C++, CUDA C/C++, OpenMP, MPI, NCCL, NVSHMEM, Java, JavaScript, Python

Applications: NVIDIA Nsight Systems, Nsight Compute, GLM/C++, Visual Studio, Android Studio, MATLAB, TensorFlow, Inventor, SolidWorks, Abaqus, ParaView

Operating Systems: Linux

References

Prof. Dr.-Ing. Benjamin Klusemann (Supervisor)
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Institute of Production Engineering and Systems
Leuphana University Lüneburg
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